

Supplementation of flaxseed oil diminishes skin sensitivity and improves skin barrier function and condition.

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BACKGROUND: Skin sensitivity is a common problem in the Western population correlated with changes of skin properties like skin barrier function, hydration and skin physiology. Skin properties can be modulated by dietary fatty acids (FA), especially poly-unsaturated FA. The present study was performed to evaluate the effect of daily supplementation with flaxseed oil and safflowerseed oil on healthy volunteers with sensitive skin. **METHODS:** The study was designed as a randomized, double-blind 12-week intervention with 2 female treatment groups (n = 13). Plasma FA profile, skin sensitivity, skin hydration, transepidermal water loss (TEWL) and skin surface were evaluated on day 0, week 6 and week 12. **RESULTS:** Supplementation with flaxseed oil led to significant decreases in sensitivity (after nicotinate irritation), TEWL, skin roughness and scaling, while smoothness and hydration were increased. Concomitantly, the ratio of n-6/n-3 FA in plasma decreased. Upon supplementation with safflowerseed oil, only a significant improvement in skin roughness and hydration was observed; however, the effects were less pronounced and determined at a later point in time than with flaxseed oil. The plasma n-6/n-3 FA ratio increased. **CONCLUSION:** The data provide evidence that daily intake of flaxseed oil modulates skin condition. Copyright © 2010 S. Karger AG, Basel.